The empirical recurrence for OEIS A299124, recovered from its tail

Adrian Perez Fontelles
Independent researcher

8 September 2026

Abstract

OEIS A299124 records "Empirical recurrence of order 66" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 71$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 69$. Its last coefficient is $c_{66} = 540 \neq 0$, so no further tail satisfies anything shorter; any order-66 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A299124 is "Number of nX4 0..1 arrays with every element equal to 0, 1, 3, 4, 5 or 7 king-move adjacent elements, with upper left element zero.". It has offset 1 and begins

$$5, 16, 23, 232, 623, 3368, 20226, 95305, \ \ldots$$

The entry states:

> Empirical recurrence of order 66 (see link above)

As of the "Last modified" line on the live entry (Feb 03 2018 08:51:45, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \;=\; \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 71$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 71$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 66: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 66.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 142$ exact terms from $n = 3$ on, it returns order 66 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{66} c_i\, a(n-i),$$

c_1	4	c_{18}	−2992414	c_{35}	−399201091	c_{52}	−54677698
c_2	8	c_{19}	10155046	c_{36}	−485433072	c_{53}	−182276009
c_3	64	c_{20}	975382	c_{37}	27727808	c_{54}	−82242688
c_4	−289	c_{21}	11775086	c_{38}	330402473	c_{55}	3118951
c_5	−457	c_{22}	−22262563	c_{39}	323033388	c_{56}	50624590
c_6	−1214	c_{23}	−3193670	c_{40}	89178256	c_{57}	45351328
c_7	7437	c_{24}	−25061700	c_{41}	−346112271	c_{58}	8951971
c_8	8386	c_{25}	26197105	c_{42}	−463117747	c_{59}	−4167730
c_9	11770	c_{26}	20511638	c_{43}	8780638	c_{60}	−8039954
c_{10}	−95069	c_{27}	25807789	c_{44}	428137258	c_{61}	−4537846
c_{11}	−73250	c_{28}	52006624	c_{45}	387678396	c_{62}	−1294307
c_{12}	−82958	c_{29}	−83032128	c_{46}	−88590853	c_{63}	−346670
c_{13}	691676	c_{30}	−123761352	c_{47}	−623831761	c_{64}	−37633
c_{14}	353495	c_{31}	−206423659	c_{48}	−324471215	c_{65}	5505
c_{15}	552530	c_{32}	241145007	c_{49}	191957256	c_{66}	540
c_{16}	−3201804	c_{33}	405686076	c_{50}	378733266		
c_{17}	−946324	c_{34}	185800472	c_{51}	228102774		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 69$, and it is the recurrence OEIS A299124 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{66} - c_1 x^{66-1} - \cdots - c_{66}$. Its constant term is $-c_{66} = -540$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 66 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 66, so $\deg q = 66$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 66, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 22 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 66, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 540.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-66 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A299124.

[2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.

[3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)

[5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.